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Distribution and potential effects of water beetles in lakes recovering from acidification

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Abstract. Regional acidification of aquatic habitats has caused severe reductions in biodiversity. Reduced sulfur dioxide emissions over the past several decades have resulted in increased pH and alkalinity in some areas of North America and Europe. However, biological recovery has lagged behind increases in lake pH. We propose that acidification-induced changes in predator assemblages can provide *biological resistance* to recovery of assemblages in lower trophic levels in lakes. Many recovering lakes remain fishless because of low colonization rates and, therefore, support a high abundance of macroinvertebrate predators that may have a large impact on zooplankton community structure. We assessed the distribution of water beetles in relation to pH and presence/absence of fish in 29 lakes on the Canadian Shield. We found that water beetle assemblage composition was not related to pH. However, the occurrence of fish was significantly negatively correlated with water beetle distribution, particularly for the predaceous diving beetle, *Graphoderus liberus*. Mesocosm experiments in Swan Lake, a fishless lake recovering from acidification, revealed that larval *G. liberus* can reduce total zooplankton abundance, species richness, and species diversity. In particular, the densities of 4 taxa (*Leptodiptomus minutus*, *Diaphanosoma birgei*, *Bosmina* [Bosmina] spp., and calanoid copepodids) were reduced by larval *G. liberus* predation. The high abundance of *G. liberus* in the absence of fish and the impact of larval *G. liberus* predation on crustacean zooplankton in Swan Lake suggest that biological resistance may be an important impediment to the recovery of aquatic food webs, despite increasing pH.

Key words: aquatic beetles, predation, pH, dytiscid larvae, biological resistance, recovery, zooplankton.

Human activities can have severe consequences for aquatic biota and are expected to continue to have large effects on biodiversity (Sala et al. 2000, Brönmark and Hansson 2002), through the direct effects of introduced toxins (e.g., Hanazato 2001), indirect effects that cascade through food webs (e.g., Forrester et al. 1999), and interactions with additional stressors (Schindler et al. 1996, Yan et al. 1996). These complex interactions have the consequence that removal of the disturbance does not always result in complete recovery of the system because of the occurrence of alternative stable states (Scheffer et al. 2001), legacy effects (Moorhead et al. 1999, Lundberg et al. 2000), and time lags.

Research on the regional acidification of aquatic habitats has provided insights into direct and indirect effects of disturbance on food webs and information about factors influencing recovery once a disturbance is removed (Yan et al. 2003). The consequences of acidification for aquatic biota are severe and include decreased population abundance or local extirpation of many species (Baker and Christensen 1991). Differences in acid sensitivity among fish species result in predictable fish assemblages along pH gradients (Rahel 1984), and at low pH (e.g., pH <6 to pH <5, depending on the species) reproductive failure (Magnuson et al. 1984) and extirpation of populations (Tammi et al. 2003) can occur. Macroinvertebrate assemblages also are shaped by acidification when acid-sensitive species are lost or reduced in abundance and acid-tolerant species flourish (McNicol et al. 1995, Carbone et al. 1998, Ledger and Hildrew 2005). Species

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richness and diversity of zooplankton are lower in acidic lakes because of the loss of acid-sensitive species (Marmorek and Korman 1993). At low pH, the zooplankton community tends to be dominated by a single acid-tolerant species, *Leptodiptomus minutus* (Lilljeborg) (Sprules 1975, Keller and Pitblado 1984). Phytoplankton communities experience similar shifts in composition related to loss of species, particularly chrysophytes, and increases in dinoflagellates at low pH (Nicholls et al. 1992).

Reductions in sulfur emissions and subsequent decreases in lake and stream sulfate concentrations in Europe and North America (Stoddard et al. 1999, Jeffries et al. 2003, Fowler et al. 2005) have been accompanied by some evidence of biological recovery (Keller et al. 1992, Battarbee 1999, Tipping et al. 2002, Gunn and Sandøy 2003, Monteith et al. 2005). However, recovery is generally considered incomplete when the community is judged against communities in nonacidified systems (e.g., Holt and Yan 2003, Yan et al. 2004). Several hypotheses for this delayed recovery have been proposed, including insufficient chemical recovery (Skjelkvåle et al. 2003, Lepori and Ormerod 2005), dispersal limitation (Snucins 2003, Binks et al. 2005), metal intolerance (Yan et al. 2004), and the biological resistance hypothesis, which postulates that interactions within the existing, acidification-structured food web may continue to shape the community despite increased pH (Binks et al. 2005, Ledger and Hildrew 2005, Frost et al. 2006). For example, invertebrate predators, such as *Chaoborus* spp. (Lichtenstein) (Diptera:Chaoboridae), thrive in fishless lakes, can consume a large proportion of the crustacean zooplankton production (Yan et al. 1991), and may impede the recovery of acid-sensitive zooplankton despite increased pH (Keller et al. 2002). This particular mechanism may be particularly relevant in lakes where fish populations have been extirpated by acidification and invertebrate predators are at the top of the aquatic food web (McNicol et al. 1995, Strong and Robinson 2004).

We sought to test the biological resistance hypothesis by asking: 1) Is water beetle community structure influenced by historical acidification or loss of fish communities? and 2) Could water beetle predation pose a barrier to recovery of other components of the food web, such as crustacean zooplankton? The relationship between acidification and community structure is well understood for many aquatic taxa (Schindler 1988, Havas and Rosseland 1995), but the influence of acidification on water beetles in permanent lentic habitats has received little attention. Water beetles include a wide range of predators (e.g., Dytiscidae and Gyrinidae), generalized algivores and

detritivores (e.g., Hydrophilidae), and consumers of filamentous algae (e.g., Haliplidae). Like many other aquatic macroinvertebrates, many water beetles are vulnerable to fish predation. Therefore, water beetles are likely to respond to changes in available food resources, such as periphyton, macrophytes, and zooplankton, as well as to changes in fish community composition. We expected that water beetle assemblages would be sensitive to direct effects of low pH or indirect effects associated with changes in the aquatic food web. Aquatic Coleoptera are seldom used in ecological or applied limnological studies, probably because of the taxonomic difficulty associated with some groups (e.g., dytiscids) and some misconceptions about their biology and ecology (Ribera and Isart 1994). However, water beetles are very common and diverse in most habitats year round and have a wide range of food preferences, environmental tolerances, and colonization ability (Foster et al. 1992, Foster 1987).

We examined the distribution of aquatic beetle assemblages in and near Killarney Park, Ontario, Canada, a region with lakes of varying pH resulting from historical acid deposition and subsequent recovery (Gunn and Sandøy 2003). The lakes ranged in pH from 4.6 to 7.4. In many of the lakes, fish populations were lost because of acid deposition originating from nearby metal smelters (Beamish and Harvey 1972). With increases in pH, some of the lakes have been restocked with fish that occurred there originally (Snucins and Shuter 1991, Snucins and Gunn 1998). However, other lakes have not had active intervention, and they remain fishless because of low colonization rates even though the pH is now high enough to support fish (Snucins and Gunn 1998). Our first study objective was to assess the distribution of water beetles in an effort to tease apart the effects of acidification from fish predation. We achieved this objective by surveying 29 lakes with and without fish along a wide range of pH.

We also were interested in determining the effect of water beetles on the recovery of lower trophic levels, specifically the crustacean zooplankton. Lake survey data may confound the effects of fish and macroinvertebrates because both fish and macroinvertebrates can have large impacts on zooplankton community structure. Therefore, we designed a manipulative field experiment to assess the potential effects of water beetle predation on zooplankton recovery. The predatory water beetle *Graphoderus liberus* (Say) (Coleoptera:Dytiscidae) can be abundant in lakes without fish, regardless of pH (Bendell and McNicol 1987). Previous studies have indicated that some dytiscid larvae can have large impacts on zooplankton communities (Arts

et al. 1981, Schum and Maly 2000). Therefore, our objectives for the experiment were to: 1) determine if larval *G. liberus* predation in a recovering lake could reduce richness and diversity by preying on crustacean zooplankton, and 2) to determine if *G. liberus* could impede recovery of acid-sensitive zooplankton.

Methods

Water beetle distribution

Field survey.—We surveyed 29 lakes during July and August of 2001 and 2002 to assess the distribution of water beetles in relation to midsummer pH and the presence of fish. Most of the lakes were small (surface area <30 ha), shallow (mean depth <12 m), and nutrient poor (Table 1). Twenty lakes were located in Killarney Wilderness Park, on the northern shore of Georgian Bay in Lake Huron. A white quartzite mountain range runs through the Park and, as a result, the buffering capacity of many of the lakes is low, making them highly susceptible to acid deposition from mining and smelting sources in nearby Sudbury. Six of the study lakes were in the City of Greater Sudbury. We had difficulty finding circum-neutral fishless lakes in the Sudbury/Killarney Park region, so 3 of the study lakes were located in Algonquin Park, Ontario, an area also on the Canadian Shield but with less acid deposition than the Sudbury area (Zeng and Hopke 1994). We attempted to sample fishless lakes and lakes with known populations of fish, including primarily *Lepomis macrochirus* (Rafinesque), *L. gibbosus* (Linnaeus), *Micropterus dolomieu* (Lacepède), and *Perca flavescens* (Mitchill) (Table 1) across the pH range. However, our lowest pH for a lake with fish was 4.9, and our highest pH for a lake without fish was 6.0.

We visited each lake once in both summers. In most lakes, we measured pH from water samples taken from the littoral area using a field pH meter (Piccolo Plus, Hanna Instruments, Ann Arbor, Michigan). Our pH estimates were similar to pH data obtained from mid-lake samples taken as part of other studies (Table 1). We sampled water beetles in the littoral area of each lake using D-net sweeps for a mean (± 1 SD) total of $\sim 3.1 \pm 0.6$ h in both years. We sampled in a variety of microhabitats, including macrophyte beds, rocky shores, organic-rich sediments, and open water. In 2002, we sampled beetles overnight using bottle traps baited with dried dog food in each lake except those with difficult access (Maggie, Patten, and Shingwak lakes). We used bottle traps to complement our sweep sampling to ensure collection of a diverse assemblage. We preserved water beetles in 70% ethanol and later identified individuals to species. We deposited vouch-

er specimens in the research collection of YA (Department of Biology, Laurentian University, Sudbury, Ontario).

Data analyses.—We calculated taxon richness of each lake as the total number of taxa observed in both summers because we were interested in the total taxon pool and expected that taxa might be absent in either of the 2 summers because of taxon turnover or low detection probabilities (Arnott et al. 1999). We conducted stepwise multiple regression analyses with pH, dissolved organic C (DOC), log(area), and log(maximum depth) as continuous independent variables, fish presence/absence as a dummy independent variable, and taxon richness as the dependent variable. Taxon richness was normally distributed (Shapiro-Wilk test, $W = 0.95$, $p = 0.24$), variances were homogeneous, as assessed by plots of residuals vs predicted values, and none of the values had exceptional influence, as assessed by Cook's D_i .

We used ordination methods to explore the relationship between environmental variables and water beetle assemblage composition. We based water beetle composition on the total number of individuals found in all samples for both years. We transformed the data using $\log(x + 1)$ for analyses to give less weight to the few abundant species. We did a preliminary Detrended Correspondance Analysis (DCA), detrended by segments, to determine if a unimodal or linear method should be used. DCA revealed gradient lengths that were <3 units (maximum length was 1.3), indicating that a linear method such as Redundancy Analysis (RDA) was appropriate (Lepš and Šmilauer 2003). Therefore, we used RDA to analyze data for species that were present in >4 study lakes, with explanatory environmental variables that included fish presence/absence, pH, lake surface area, mean lake depth, maximum lake depth, total P concentration, and DOC concentration (Table 1). We were interested primarily in the influence of fish and pH on water beetle assemblages, so we also ran an RDA with only fish presence/absence and pH as explanatory variables.

Mesocosm experiment

Experimental design.—We conducted an experimental mesocosm study in Swan Lake, Sudbury, to determine the impact of *G. liberus* on recovering crustacean zooplankton communities. Swan Lake is a small (surface area = 5.8 ha, maximum depth = 8.8 m) lake that was historically acidified by local sulfur deposition. The lake is currently fishless, but a paleolimnological study using *Chaoborus* remains indicates that it once contained fish (Uutala and Smol 1996). Both the

TABLE 1. Chemical and physical characteristics of the 29 study lakes in the Canadian Shield area. All data were obtained from Snucins and Gunn (1998) except as noted. P = present, A = absent, NA = not available, TP = total P, DOC = dissolved organic C.

Lake	Latitude	Longitude	Fish	pH	TP (µg/L)	Surface area (ha)	Maximum depth (m)	Mean depth (m)	DOC (mg/L)
No. 64	46°02'08"N	81°24'49"W	A	5.3	NA	3.6	2.5	0.7	3.4
No. 65	46°01'57"N	81°24'52"W	A	5.5	NA	2.6	6.4	2.1	2.6
Acid	46°02'01"N	81°26'38"W	A	5.0	2.0	19.6	29.0	10.9	1.6
Bat ^a	45°08'55"N	78°37'57"W	A	4.6	14.0	2.3	8.5	2.9	4.3
Heaven	46°04'42"N	81°17'38"W	A	4.8	6.0	1.7	17.8	5.5	4.0
Hemlock	46°05'00"N	81°17'06"W	A	4.7	6.0	3.3	4.5	2.6	1.2
Lumsden	46°01'23"N	81°25'59"W	A	5.2	6.0	23.8	21.8	9.0	1.5
Maggie ^a	45°30'00"N	78°52'00"W	A	5.8	5.4	139.0	28.0	10.2	1.6
Nick ^b	45°18'54"N	78°18'05"W	A	6.0	21.0	2.3	6.5	3.0	NA
Shingwak	46°04'28"N	81°19'06"W	A	4.7	2.0	5.3	21.8	9.6	0.3
Swan ^c	46°21'51"N	81°03'56"W	A	5.5	8.1	5.8	8.8	2.8	2.0
Whiskeyjack	46°04'57"N	81°17'29"W	A	4.6	2.0	12.8	42.7	19.2	0.4
Artist	46°03'13"N	81°26'58"W	P	5.7	NA	26.0	1.5	1.0	2.0
A.Y. Jackson	46°01'17"N	81°23'57"W	P	5.8	4.0	6.5	9.8	7.0	2.7
Clearwater ^c	46°22'05"N	81°03'04"W	P	6.2	0.1	76.0	21.5	8.4	2.7
Grey	46°07'46"N	81°10'14"W	P	4.9	6.0	31.8	11.8	5.4	3.4
Grow	46°10'05"N	81°27'17"W	P	6.6	6.0	13.1	9.0	5.3	4.1
Hannah ^c	46°11'02"N	81°33'55"W	P	7.4	5.7	27.3	8.5	4.0	3.3
Log Boom	46°07'07"N	81°14'14"W	P	5.5	6.0	6.9	5.5	3.4	2.4
Lohi ^c	46°23'05"N	81°02'36"W	P	6.4	3.9	40.5	19.5	6.2	1.2
Middle ^c	46°26'10"N	81°01'32"W	P	7.1	5.2	28.2	15.0	6.2	3.0
Muriel	46°03'04"N	81°26'11"W	P	5.1	2.0	31.7	12.2	6.0	1.2
Partridge	46°04'58"N	81°18'12"W	P	5.7	2.0	11.0	16.9	6.2	1.8
Patten	46°06'36"N	81°21'37"W	P	5.1	4.0	11.9	6.4	2.2	3.5
Sans Chambre ^c	46°43'08"N	81°07'53"W	P	6.2	8.0	14.5	15.0	5.6	3.0
Teardrop ^d	46°02'33"N	81°24'48"W	A	6.5	6.0	3.4	16.6	9.6	1.0
Terry	46°03'56"N	81°17'19"W	P	5.4	6.0	11.5	8.0	3.1	5.5
Turbid	46°06'49"N	81°11'23"W	P	5.0	6.0	20.7	9.1	3.8	3.2
Lake of the Woods	46°06'09"N	81°12'10"W	P	4.9	8.0	9.7	6.0	2.5	3.3

^a Dorset Environmental Science Centre, Ontario Ministry of the Environment, unpublished data

^b Malkin et al. (2006)

^c B. Keller, Ontario Ministry of the Environment, unpublished data

^d Contains lake trout and sculpins, which inhabit deep, cold waters; fish predation absent in the littoral regions during the summer when the lake is thermally stratified

pelagic and littoral regions of the lake support high numbers of macroinvertebrate predators, including *G. liberus*, *Notonecta*, *C. americanus*, and *C. puntipennis* (SEA, personal observations). Water-chemistry records indicate a recovery of pH from <4.0 in 1977 to an average pH of 5.6 (range: 5.2–6.1) in 2001 (Cooperative Freshwater Ecology Unit, Laurentian University, unpublished data). Despite improvements in water chemistry, the crustacean zooplankton community consists mostly of a single, acid-tolerant species *Leptodiaptomus minutus* (>90% total crustacean zooplankton biomass) and lower densities of *Bosmina* (*Bosmina*) spp. (Müller) and *Diaphanosoma birgei* (Arnott et al. 2001).

We established 2 treatments, each with 4 replicates: 1) control—with no predators, and 2) predator—with predatory *G. liberus* larvae. The crustacean zooplankton community consisted primarily of Swan Lake

zooplankton, stocked at ambient densities. In addition, we added a small number (13.3% ambient density) of zooplankton colonists from Kelly Lake (lat 46°26'N, long 81°05'W), a nearby lake with a diverse zooplankton assemblage, to each enclosure. We chose Kelly Lake as a source of colonists because its zooplankton community is quite diverse and because it is a relatively large lake, located within 6 km of Swan Lake. Thus, we reasoned that Kelly Lake could be a potential source of overland dispersers. Kelly Lake taxa included *Daphnia* spp., *Bosmina* (*Bosmina*) spp., *Ceriodaphnia lacustris* (Birge), *Chydorus* spp. (Leach), *D. birgei*, *Acanthocyclops vernalis* complex (Fischer), *Acroperus harpae* (Baird), *Diacyclops bicuspidatus thomasi* (Forbes), *Mesocyclops edax* (Forbes), *Cyclops scutifer* (Sars), *Tropocyclops extensus* (Kiefer), *L. minutus*, and *Skistodiaptomus oregonensis* (Lilljeborg). These colonists were added to determine if the presence of *G. liberus*

could influence the successful establishment of zooplankton colonists.

We set up experimental mesocosms, consisting of cylindrical polyethylene bags (Filmtech Plastic, Brampton, Ontario), 1 m in diameter and 1.9 m deep, suspended from wood and Styrofoam frames that were oriented in a north-south direction in Swan Lake. The maximum depth of Swan Lake is 8.8 m, but ~67% of the total lake volume occurs at depths <2.5 m from the surface. Therefore, the bags were representative of most of the habitat available to zooplankton in Swan Lake. Bags were closed at the bottom and open at the surface to allow exchange of gases with the atmosphere. The surface opening of each bag was covered with mesh to prevent aerial colonization by other macroinvertebrates.

On 20 June 2001, we filled enclosures with Swan Lake water that had been filtered through 60- μ m mesh to remove zooplankton. On 21 and 22 June, we stocked enclosures with Swan and Kelly lake zooplankton. In both lakes, we collected zooplankton from the top 2 m of the water column using a 25-cm-diameter, 80- μ m-mesh conical net. We added the zooplankton captured in 16 vertical hauls from Swan Lake and 2 vertical hauls from Kelly Lake to each enclosure. We removed mites and *Chaoborus* larvae from the inoculum prior to stocking the enclosures. On 22 June, we collected *G. liberus* from the littoral area of Swan Lake using small kitchen sieves. We stocked 35 *G. liberus* larvae in each predator-treatment enclosure: 18 first-instar, 7 second-instar, and 10 third-instar larvae. On 10 July, we restocked 5 first-instar and 2 third-instar *G. liberus* larvae into the predator treatment enclosures to replace individuals that had died or metamorphosed. We did not restock 2nd-instar larvae because we assumed that many of the previously stocked 1st-instar larvae had metamorphosed into 2nd-instar larvae. The density of *G. liberus* larvae in the enclosures was similar to the density observed in Swan Lake, as determined visually.

Starting on 25 June, we sampled crustacean zooplankton in the enclosures weekly for 5 wk using a 14-cm-diameter, 80- μ m-mesh conical net. We combined 2 net hauls from 1.2 m and preserved the sample in 4% sugar-formalin solution for later identification and counting. We identified crustacean zooplankton using a Leica MZ12.5 stereomicroscope (Leica Microsystems, Richmond Hill, Ontario, Canada). We examined several subsamples, such that a total of ~400 individuals were counted per total sample and that no species or genus was >20% of the total count. We identified adults to either species or genus, and juvenile copepods as either calanoid or cyclopoid.

Data analyses.—We defined taxon richness for each

treatment as the cumulative number of genera or species in each replicate over the entire experiment, excluding immatures. The Shannon-Wiener diversity index was calculated as:

$$H' = -\sum p_i \log(p_i)$$

where p_i is the proportion of the total abundance consisting of taxon i .

We analyzed the impact of *G. liberus* predation on taxon richness and diversity using repeated measures 1-way analysis of variance (RM ANOVA). We analyzed differences in taxon abundance between treatments using a multiple repeated-measures analysis of variance (RM MANOVA) for 13 zooplankton taxa that were detected in the experimental enclosures. We used all samples, including the initial sample taken 3 d after predators were added, in the analyses. We omitted 4 zooplankton taxa because their abundances were low, and they were not reliably detected. We detected heteroscedasticity of variances with a Cochran's and Bartlett-Box test. Therefore, we $\log(x + 1)$ transformed all zooplankton abundances. We analyzed individual taxa separately using a 1-way RM ANOVA. We adjusted α using the sequential Bonferroni method to account for the large number of statistical tests.

We calculated prey selectivity for 8 dominant zooplankton taxa using Manly's Alpha (Krebs 1989):

$$\alpha_i = (r_i/n_i)(1/\sum(r_j/n_j))$$

where r_i , r_j , is the proportion of prey type i or j in the diet (mean % difference in density of each zooplankton taxon between predator and control enclosures) and n_i , n_j is the proportion of prey type i or j in the environment (mean proportional representation of each zooplankton taxon in control enclosures).

Results

Lake survey

We collected 2553 water beetles from our 29 study lakes. We identified 72 species belonging to the families Dytiscidae, Gyrinidae, Haliplidae, and Hydrophilidae (Appendix). Most species were rare in terms of the number of individuals collected and the number of lakes in which they were found (Appendix, Fig. 1). Together, the 50 rarest species made up <10% of the total number of individuals collected, and only 5 individual species contributed >5% of the numerical total (Appendix). The mean number of species found per lake was 14. Maggie Lake, a fishless lake in a region that was not heavily affected by acid rain, had the fewest observed species (4), whereas Hannah Lake, a circumneutral lake with fish, had the most observed

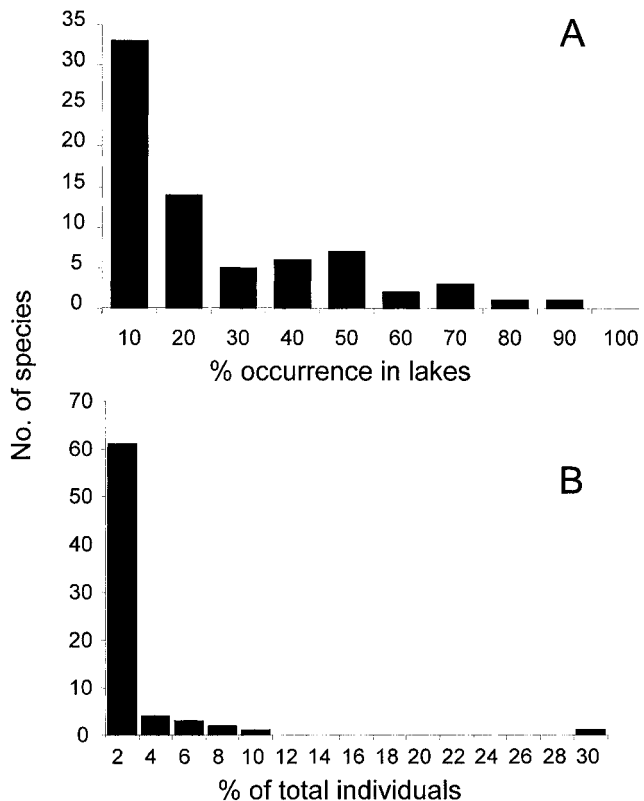


FIG. 1. Frequency distributions of the number of water beetle species in 29 study lakes in the Canadian Shield area (A) and of the number of individuals per species across the 29 lakes (B).

species (27). Beta diversity (total number of species collected/mean lake taxon richness) was 5.0, indicating that spatial turnover was high.

Water beetle richness was significantly related to pH ($R^2 = 0.18$, $p = 0.02$; Fig. 2). However, all other variables included in the stepwise multiple regression model ($\log[\text{area}]$, $\log[\text{mean maximum depth}]$, DOC, and fish presence/absence) were not statistically significant and, therefore, were eliminated from the model.

In general, the species composition of water beetle assemblages showed little influence of the environmental variables that we measured. When we used forward stepwise selection of the 7 environmental variables in the RDA, only fish presence/absence explained a significant amount of variation in water beetle composition (8%, $p = 0.002$). Lake mean depth explained an additional 5% of the variation, but this effect was not statistically significant ($p = 0.08$) using a Monte Carlo permutation test with 999 randomizations. The distribution of one species, *G. liberus*, was negatively associated with fish, and fish presence/absence explained >65% of its variation in abundance (Figs 3, 4).

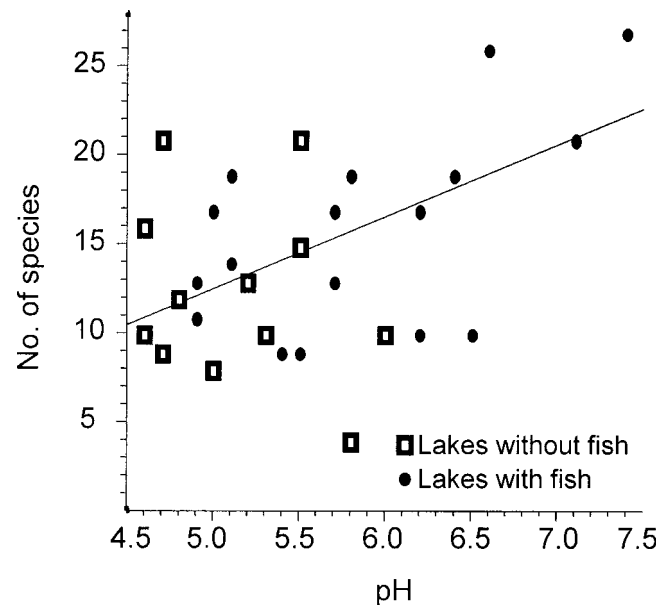


FIG. 2. Relationship between water beetle taxon richness and lake water pH in 29 study lakes in the Canadian Shield area.

Mesocosm experiment

The mesocosm experiment revealed a significant impact of *G. liberus* on crustacean zooplankton. Cumulative taxon richness and Shannon–Wiener diversity were significantly lower in the predator treatment than in the control treatment (Table 2). Predation by *G. liberus* reduced the total abundance of zooplankton by 10 ind./L (21% reduction). This reduction was primarily a consequence of effects on 4 taxa: *Bosmina* (*Bosmina*) spp., *L. minutus*, *D. birgei*, and calanoid copepodids, which had lower densities in the predator treatment than in the control treatment (Tables 2, 3; Fig. 5). Juvenile copepods (cyclopoid copepodids, cyclopoid nauplii, and calanoid nauplii) had higher densities in the predator treatment than in the control treatment. Many of the colonist species from Kelly Lake (*Daphnia* spp., *D. b. thomasi*, *C. lacustris*, *A. vernalis* complex, and *M. edax*) did not thrive in our mesocosms. The average density of these taxa throughout the experiment was <1 ind./L, and all species except *D. b. thomasi*, which was found in 50% of the enclosures, occurred in ≤ 1 enclosure at the end of the experiment. Three of the colonist taxa, *Daphnia* spp., *C. lacustris*, and *M. edax*, were detected in a least one of the control enclosures, but not the predator enclosures, at the end of the experiment.

The calculated selectivity indices for each taxon generally agreed with the changes in abundance associated with predator treatments. *Graphoderus liberus* tended to select larger prey items, including

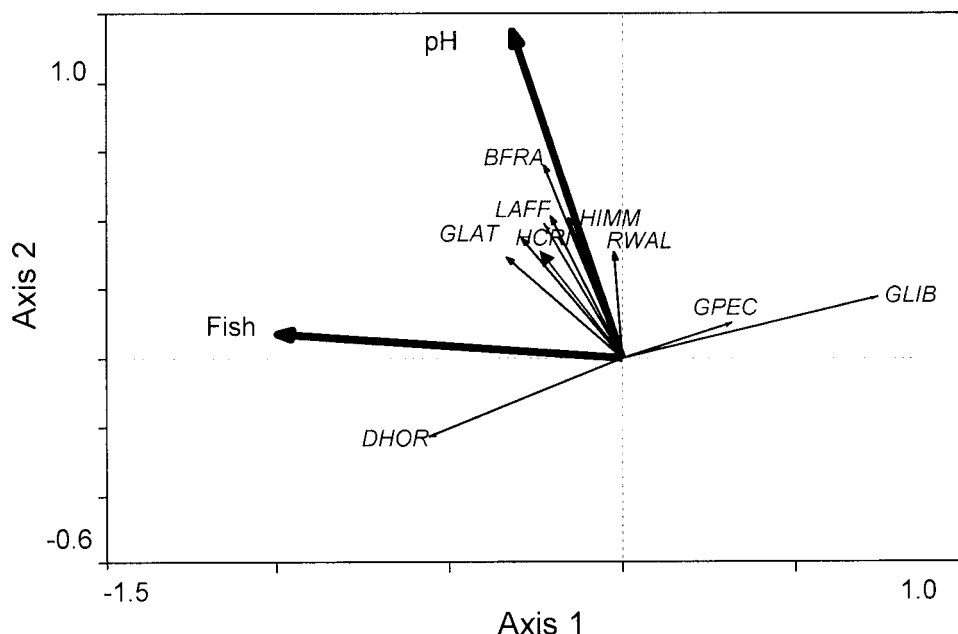


FIG. 3. Redundancy analysis for water beetle assemblages using pH and fish presence/absence as explanatory variables. Species with >10% of their variation explained by either axis are shown. See Appendix for species abbreviations.

D. birgei and *L. minutus*, and tended to avoid the smallest prey items, juvenile copepods and *Chydorus* spp. (Table 3). Although *Bosmina* are intermediate in size, they may have been selected because they were the most abundant taxon in the community.

Discussion

Relationship between pH and water beetle community structure

We observed a weak relationship between pH and water beetle richness (i.e., only 18% of the variation in richness was explained by pH), and we failed to detect a relationship between lake pH and water beetle composition for the 29 lakes surveyed, despite the relatively large pH gradient across lakes (pH range: 4.6–7.4). However, we did find a significant, although weak, effect of fish on water beetle composition. A small amount of the variation in water beetle composition (8%) was explained by the presence/absence of fish, which was the direct result of acidification in many lakes in Killarney Park and Sudbury. In particular, the predatory dytiscid, *G. liberus*, was abundant in the absence of fish and occurred across a wide range of pH.

Other studies in areas affected by regional acidification have found similar results for benthic and littoral macroinvertebrates (but see Friday 1987). Some mayflies (Carbone et al. 1998, Snucins 2003), chironomids (Ledger and Hildrew 2005), and water beetle species

(Alarie and Leclair 1988, Cuppen 1986, Juliano 1991) are sensitive to pH, but the distribution and abundance of many aquatic insects seems to be driven primarily by the occurrence of fish predators (McNicol and Wayland 1992, Bendell and McNicol 1995, Welbourn et al. 1996). Synoptic surveys aimed specifically at water beetles have reported similar trends. Foster (1995) found no detectable effect of pH on water beetles in an experimental liming of a bog. Beetle assemblage structure was strongly related to the

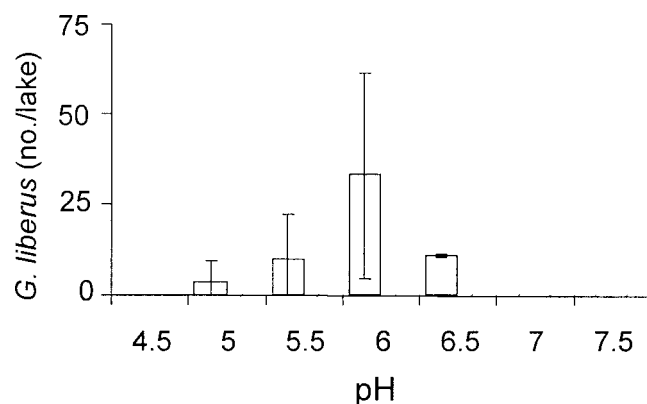


FIG. 4. Mean (± 1 SE) number of *Graphoderus liberus* caught per lake in 29 study lakes in the Canadian Shield area. Open bars indicate lakes without fish. Solid bars indicating lakes with fish are not visible at this scale because only one beetle was found in all of the lakes with fish.

TABLE 2. Results from repeated measures one-way analysis of variance of data from the mesocosm experiment. * indicates statistical significance at $\alpha = 0.05$, ** indicates statistical significance after sequential Bonferroni adjustment of α .

Response variable	Predator effect		Predator $\times$ time effect	
	F	p	F	p
Cumulative taxon richness	19.69	0.004**	1.44	0.251
Shannon–Weiner diversity	32.66	0.001**	14.47	<0.001**
<i>Leptodiptomus minutus</i>	201.48	<0.001**	2.28	0.09
Calanoid copepodid	1.72	0.237	3.48	0.022*
Calanoid nauplii	0.104	0.76	4.39	0.008*
<i>Ceriodaphnia lacustris</i>	0.17	0.70	0.68	0.61
<i>Chydorus</i> spp.	0.07	0.796	0.51	0.73
<i>Diacyclops bicuspidatus thomasi</i>	3.76	0.101	1.79	0.16
Cyclopoid copepodid	0.06	0.81	3.21	0.03*
Cyclopoid nauplii	1.18	0.32	6.5	0.001**
<i>Diaphanosoma birgei</i>	9.39	0.022*	4.3	0.009*
<i>Bosmina</i> (<i>Bosmina</i>) spp.	17.86	0.006**	0.56	0.70

presence of fish in other studies (Bendell and McNicol 1995, Fairchild et al. 2000, Tate and Hershey 2003). In the absence of fish, beetle biomass is high, individual sizes are large and, as found in our study, pelagic larvae such as those of the dytiscid species *G. liberus* are abundant (Fairchild et al. 2000).

The response of aquatic food webs to pH is often linked to the reduction or elimination of fish predators (Eriksson et al. 1980). At low pH, fish reproduction is impaired and, if the duration of stress is long enough, species are lost, resulting in reduced richness or the complete absence of fish (Minns 1989, Matuszek et al. 1990). The elimination of fish predators enables macroinvertebrate predators to flourish. The continued absence of fish continues to shape invertebrate community structure, even after the pH of the lake has increased (Eriksson et al. 1980). In Killarney Park, where historical pH was <5.0 in many lakes (Sprules 1975), fish populations were lost (Beamish and Harvey 1972). In many lakes, all fish species were eliminated (Snucins and Gunn 1998). Acid-tolerant macroinvertebrates species that are susceptible to fish predation probably flourished (Bendell and McNicol 1987). Predatory dytiscids are early colonists of new habitats (Fairchild et al. 2000) and probably colonized fishless sites while pH was low. In particular, *G. liberus* appears to have maintained high population abundance as pH has increased with recovery in the lakes that remained fishless. Our data suggest that *G. liberus* can tolerate a wide range of pH but is eliminated when fish are present (Fig. 4).

TABLE 3. Mean (± 1 SD) abundances of crustacean zooplankton in control enclosures (averaged over 4 wk) and % difference in density of individual taxa between predator and control treatments. Manly's Alpha indicates prey preference. Values >0.11 (1/n) indicate prey that were selected more than expected assuming predators sampled prey randomly; values <0.11 indicate prey that were selected less than expected assuming predators sampled prey randomly.

Taxon	Mean abundance (ind./L)	% difference	Manly's Alpha
<i>Diaphanosoma birgei</i>	4.8 (1.5)	-58	0.21 (0.05)
<i>Leptodiptomus minutus</i>	7.6 (2.2)	-83	0.32 (0.12)
<i>Bosmina</i> (<i>Bosmina</i>) spp.	10.5 (3.3)	-50	0.22 (0.16)
<i>Ceriodaphnia lacustris</i>	0.09 (0.09)	+16	0.10 (0.11)
Cyclopoid copepodids	0.9 (0.2)	+42	0.02 (0.03)
Calanoid copepodids	8.4 (0.6)	-17	0.07 (0.08)
Cyclopoid nauplii	6.9 (1.6)	+45	0.01 (0.02)
Calanoid nauplii	9.2 (2.3)	+13	0.03 (0.03)
<i>Chydorus</i> spp.	2.3 (1.5)	+31	0.03 (0.06)

Effect of water beetle predation on zooplankton community structure

Predation by *G. liberus* may impede recovery of other trophic levels (i.e., crustacean zooplankton) in historically acidified lakes. Our mesocosm study indicated that *G. liberus* can have a significant impact on crustacean zooplankton community structure, resulting in lowered richness, diversity, and densities of the dominant zooplankton species. This result suggests that, in recovering lakes, historical acidification can continue to manifest itself through indirect foodweb effects. Overall, predation by *G. liberus* reduced the total crustacean zooplankton concentration by 21%, primarily through predation on *D. birgei* and *L. minutus*. Many of the colonist species from Kelly Lake did not thrive in our enclosures. These species may have been unable to establish populations because of low pH or interactions with the resident crustacean zooplankton community (Binks et al. 2005). Three of our colonist taxa, *Daphnia* spp., *C. lacustris*, and *M. edax*, were detected in at least one control enclosure but not in any of the predator enclosures, suggesting that *G. liberus* may have had a predation impact on these taxa. However, we cannot draw any conclusions about the effect of *G. liberus* on colonization success of these taxa because overall densities were too low. We expect the impact of *G. liberus* predation would be high on many of the colonizing species, given their large size and the apparent preference of *G. liberus* for large taxa.

Few studies have investigated the influence of water beetle predation on zooplankton communities, despite

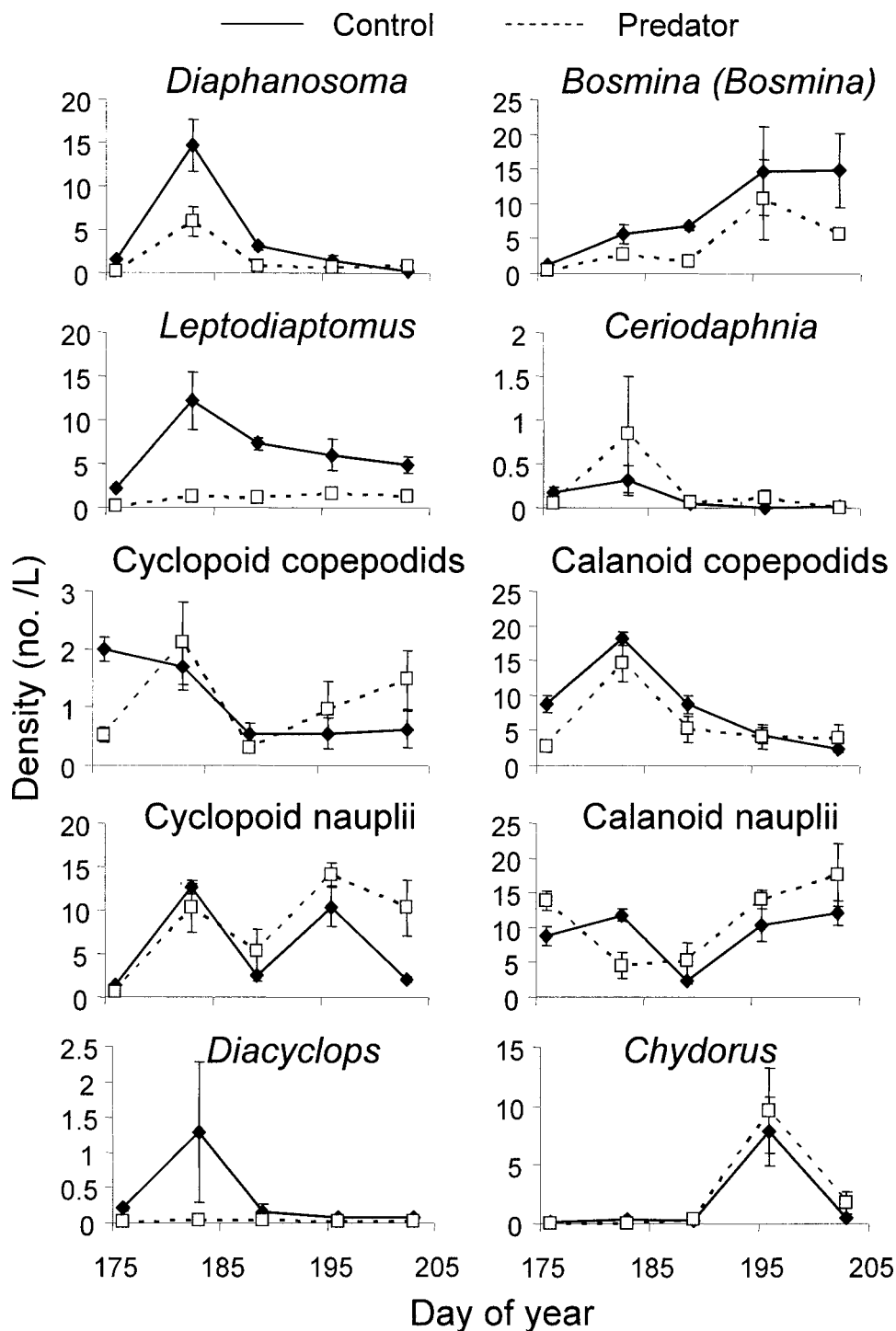


FIG. 5. Mean (± 1 SE) density of individual zooplankton taxa in the control and predator treatments of the mesocosm experiment.

the frequent occurrence of predatory water beetles in fishless lakes (Bendell and McNicol 1987). Predatory dytiscid larvae of the tribe Aciliini (e.g., species of the genera *Graphoderus* Dejean and *Acilius* Leach) are common in the pelagic area of small fishless lakes

(SEA, personal observation) and can have high feeding rates on zooplankton in experimental situations (Arts et al. 1981, Cooper et al. 1985). Nevertheless, most studies investigating the influence of macroinvertebrate predators on zooplankton have concentrated on

notonectids (e.g., Murdoch et al. 1984, Hampton et al. 2000) and larvae of Chaoboridae (e.g., *Chaoborus*) (Neill 1981, Arnett and Vanni 1993, Yan et al. 1991), and only a few studies have been directed at predatory water beetle larvae (e.g., Arts et al. 1981, Hansson and Tranvik 1996, Schum and Maly 2000). Experiments on simple Antarctic aquatic ecosystems found that individuals of the dytiscid beetle *Lancetes angusticollis* (Curtis) (treated as *L. clausi* [Müller]), consumed 3.1 *Pseudoboeckella* (Mrázek) (a copepod) per day (Hansson and Tranvik 1996). This feeding rate is low compared to the potential impact of fish, but a study that compared water beetle predation of a nondetermined species of the genus *Acilius* with another invertebrate predator (*Chaoborus americanus* [Johannsen]) found that *Acilius* sp. consumed 16× more *Daphnia* than did *Chaoborus* (Schum and Maly 2000). In our mesocosm experiment, we found that *G. liberus* had an effect on total abundance, richness, and diversity of a zooplankton assemblage. The individual taxa that were negatively affected experienced reductions in total abundance ranging from 50 to 83% (Table 3), which indicates that *G. liberus* predation may have a large impact on other zooplankton communities in fishless lakes.

High predation rates by macroinvertebrate predators such as *G. liberus* may have implications for littoral–pelagic coupling. Habitat boundaries in lakes may be blurred by interhabitat omnivory by fish that obtain a high proportion of their diet from benthic sources as well as from pelagic sources (Schindler and Scheuerell 2002). In fishless lakes, macroinvertebrate predators, typically assumed to reside in littoral areas, also may contribute to littoral–pelagic habitat coupling by preying on pelagic-based zooplankton prey. Macroinvertebrate predators often are found in the pelagic regions of small, fishless lakes, particularly at night (Hampton and Duggan 2003, SEA, personal observation), and their removal can affect zooplankton size structure, presumably through size-selective predation (Herwig and Schindler 1996). Pelagic zooplankton may show behavioral responses to littoral invertebrate predators (e.g., Dodson and Havel 1988, Lüning 1992, Van de Meutter et al. 2005), suggesting that pelagic zooplankton are vulnerable to predation by littoral invertebrate predators. Our study builds evidence for this premise by demonstrating that a dytiscid water beetle, *G. liberus*, can reduce the abundance of pelagic taxa by >20% and can decrease crustacean zooplankton richness and diversity.

One of the criticisms of mesocosm experiments is that they do not adequately represent the lake in its entirety; i.e., they do not allow water–sediment–surface exchange, large-scale mixing and nutrient recycling, or

complete predator assemblages (Carpenter 1996). However, mesocosm experiments excel in determining potential mechanisms in a highly controlled environment that is subject to normal temperature, light, and nutrient conditions of the lake. Our shallow mesocosms probably did not accurately reflect interactions in the deepest part of the lake (maximum lake depth = 8.5 m), but they probably were a reasonably realistic representation of the 67% of the lake volume that was above a 2.5 m depth. Our mesocosms restricted the movement of predators, confining them to ~1500 L of water. The restricted access to littoral habitats may have resulted in an overestimate of predation impact on the pelagic species stocked in our mesocosms, but we note that there was no effect of *G. liberus* predation on *Chydorus* sp., a taxon that typically is considered littoral but was present in our mesocosms.

The impact of *G. liberus* predation on individual taxa in our mesocosm zooplankton communities was quite variable. Densities of *Bosmina* (*Bosmina*) spp., *D. birgei*, and adult *L. minutus* were reduced in predator treatments by 50% (5.3 ind./L), 58% (2.8 ind./L), and 83% (6.3 ind./L), respectively. In contrast, juvenile copepods increased from 13 to 45% in the presence of *G. liberus*. Our calculation of prey selectivity corresponded with this variable effect, suggesting that *G. liberus* prefers larger prey items. Another pelagic-dwelling larval diving beetle, a Californian species of the genus *Acilius* (most likely *A. abbreviatus* Mannerheim, the only species of *Acilius* recorded in California), preferentially consumes larger prey, and their predation rates are lower on slow-moving prey than on fast-moving prey because of reduced encounter rates (Cooper et al. 1985). In experimental trials, Cooper et al. (1985) found that swimming speeds for *Diaptomus* spp. and *Daphnia* spp. were generally faster than swimming speeds for small cladocerans and cyclopoid copepods. Given this result, it is not surprising that selectivity was highest for adult *L. minutus* and *D. birgei* in our experiments. *Daphnia* did not thrive in our experimental mesocosms, probably because of acid stress (Binks et al. 2005), but we expect that they would experience high predation rates under conditions of higher pH. In laboratory experiments, 3rd-instar *G. liberus* reduced the abundance of *Daphnia* spp. when fed a mixture of species from Kelly Lake but had little impact on other species (Jackson 2004). This result could have important ramifications for aquatic ecosystems recovering from acidification and suggests that full recovery may be limited while fish remain absent. Further investigation is merited in hopes of understanding how biological resistance by invertebrate predators can impede crustacean zooplankton recovery.

Water chemistry has improved in many lakes that were anthropogenically acidified (Keller et al. 2003), but biological recovery has lagged behind chemical recovery (Arnott et al. 2001, Yan et al. 2004). Several reasons for this lag have been suggested, including low dispersal rates, inappropriate abiotic conditions, and species interactions (Yan et al. 2003). Theoretical studies corroborate the hypothesis that communities that lose species may experience a structural reorganization that precludes the reintroduction of lost species (Lundberg et al. 2000). Our survey provides evidence that macroinvertebrate predators such as *G. liberus* thrive in the absence of fish, regardless of pH. Our mesocosm study suggests that these predators may play an important role in structuring recovering crustacean zooplankton communities. Ultimate recovery of aquatic food webs (i.e., a reduction in water beetle predators and subsequent recovery of crustacean zooplankton communities) may require active intervention to re-establish natural fish communities in addition to improvements in water quality.

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APPENDIX. Species found in survey of 29 lakes in the Canadian Shield area, species and family codes used in the redundancy analysis (1 = Dytiscidae, 2 = Gyrinidae, 3 = Halophilidae, 4 = Hydrophilidae), the % of lakes in which each species was detected (% occurrence), and the % of the total number of individuals collected in each species. Species are ordered according to their % occurrence in the survey lakes.

Species	Species code	% occurrence	% total individuals	Species	Species code	% occurrence	% total individuals
<i>Neoporus undulatus</i> (Say)	NUND1	86.2	28.2	<i>Gyrinus bifarius</i> (Fall)	GBIF2	10.3	0.4
<i>Dineutus nigrior</i> (Roberts)	DNIG2	79.3	7.1	<i>Hydroporus notabilis</i> (LeConte)	HNOT1	10.3	0.2
<i>Laccophilus maculosus</i> (Say)	LMAC1	69.0	8.2	<i>Enochrus hamiltoni</i> (Horn)	EHAM4	10.3	0.2
<i>Gyrinus pugnionis</i> (Fall)	GPUG2	65.5	6.2	<i>Rhantus binotatus</i> (Harris)	RBIN1	10.3	0.1
<i>Ilybius biguttulus</i> (Germar)	IBIG1	62.1	4.3	<i>Haliphus pantherinus</i> (Aubé)	HPAN3	6.9	0.9
<i>Tropisternus natator</i> (d'Orchymont)	TNAT4	55.2	3.4	<i>Gyrinus ventralis</i> (Kirby)	GVEN2	6.9	0.4
<i>Coptotomus longulus</i> (LeConte)	CLEN1	55.2	2.9	<i>Anacaena limbata</i> (Fabricius)	ALIM4	6.9	0.2
<i>Uvarus granarius</i> (Aubé)	UGRA1	48.3	3.3	<i>Peltodytes tortulosus</i> (Roberts)	PTORT3	6.9	0.2
<i>Enochrus ochraceus</i> (Melsheimer)	EOCH4	48.3	2.1	<i>Gyrinus dichrous</i> (LeConte)	GDIC2	6.9	0.1
<i>Gyrinus latilimbus</i> (Fall)	GLAT2	44.8	1.6	<i>Hydrochara obtusata</i> (Say)	HOBT4	6.9	0.1
<i>Tropisternus mixtus</i> (LeConte)	TMIX4	44.8	1.1	<i>Hydrocolus paugus</i> (Fall)	HPAU1	6.9	0.1
<i>Graphoderus liberus</i> (Say)	GLIB1	41.4	5.5	<i>Hydroporus striola</i> (Gyllenhal)	HSTR1	6.9	0.1
<i>Haliphus immaculicollis</i> (Harris)	HIMM3	41.4	4.2	<i>Enochrus cinctus</i> (Say)	ECIC4	6.9	0.1
<i>Enochrus consurtus</i> (Green)	ECON4	41.4	1.5	<i>Haliphus connexus</i> (Matheson)	HCON3	6.9	0.1
<i>Dineutus horni</i> (Roberts)	DHOR2	34.5	1.5	<i>Gyrinus gibber</i> (LeConte)	GGIB2	3.4	0.3
<i>Gyrinus sayi</i> (Aubé)	GSAY2	34.5	1.4	<i>Hydaticus aruspex</i> (Clark)	HARU1	3.4	0.3
<i>Hygrotus sayi</i> (Balfour-Browne)	HGSA1	34.5	1.2	<i>Hydrobius fuscipes</i> (Linné)	HFUS4	3.4	0.1
<i>Enochrus perplexus</i> (LeConte)	EPER4	34.5	0.5	<i>Haliphus leoparidus</i> (Roberts)	HLEO3	3.4	0.1
<i>Ilybius pleuriticus</i> (LeConte)	IPLE1	31.0	1.9	<i>Laccobius agilis</i> (Randall)	LAGI4	3.4	0.1
<i>Desmopachria convexa</i> (Aubé)	DCON1	31.0	1.3	<i>Agabus ambiguus</i> (Say)	AAMB1	3.4	0.0
<i>Neoporus carolinus</i> (Fall)	NCAR1	27.6	0.7	<i>Acilius medius</i> (Say)	AMED1	3.4	0.0
<i>Gyrinus pectoralis</i> (LeConte)	GPEC2	24.1	1.1	<i>Agabus seriatus</i> (Say)	ASER1	3.4	0.0
<i>Agabus anthracinus</i> (Mannerheim)	AANT1	24.1	0.7	<i>Acilus sylvanus</i> (Hilsenhoff)	ASYL1	3.4	0.0
<i>Gyrinus lecontei</i> (Fall)	GLEC2	20.7	0.5	<i>Berosus hatchi</i> (D. C. Miller)	BHAT4	3.4	0.0
<i>Hydrobius maelenus</i> (Germar)	HMAE4	20.7	0.5	<i>Berosus perigrinus</i> (Herbst)	BPER4	3.4	0.0
<i>Haliphus cribarius</i> (LeConte)	HCRI3	17.2	0.7	<i>Colymbetes sculptilis</i> (Harris)	CSCU1	3.4	0.0
<i>Liodessus affinis</i> (Say)	LAFF1	17.2	0.5	<i>Dytiscus harisii</i> (Kirby)	DHAR1	3.4	0.0
<i>Berosus fraternus</i> (LeConte)	BFRA4	17.2	0.5	<i>Dytiscus verticalis</i> (Say)	DVER1	3.4	0.0
<i>Hydroporus dentellus</i> (Fall)	HDEN1	17.2	0.4	<i>Gyrinus impressicollis</i> (Kirby)	GIMP2	3.4	0.0
<i>Paracymus subcupreus</i> (Say)	PSUB4	17.2	0.4	<i>Graphoderus occidentalis</i> (Horn)	GOCC1	3.4	0.0
<i>Acilius semisulcatus</i> (Aubé)	ASEM1	17.2	0.2	<i>Gyrinus wallisi</i> (Fall)	GWAL2	3.4	0.0
<i>Rhantus wallisi</i> (Hatch)	RWAL1	17.2	0.2	<i>Hygrotus farctus</i> (LeConte)	HGFA1	3.4	0.0
<i>Gyrinus affinis</i> (Aubé)	GAFF2	13.8	0.4	<i>Hygrotus picatus</i> (Kirby)	HGPI1	3.4	0.0
<i>Peltodytes edentulus</i> (LeConte)	PEDE3	13.8	0.4	<i>Hygrotus laccophilinus</i> (LeConte)	HLAC1	3.4	0.0
<i>Liodessus fuscatus</i> (Crotch)	LFUS1	10.3	0.5	<i>Hydroporus melsheimeri</i> (Fall)	HMEL1	3.4	0.0
				<i>Hydrocus rufipes</i> (Melsheimer)	HRUF4	3.4	0.0
				<i>Ilybius ignarus</i> (LeConte)	IIGN1	3.4	0.0